Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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Biographical Sketch

Dr. Theodore P. Pavlic

(a) Professional Preparation

School of Life Sciences, Arizona State University, e-mail: tpavlic@asu.edu, tel: +1-760-483-3390

The Ohio State University, Columbus, OH; Electrical & Computer Engineering; B.S.E.C.E., 2004
The Ohio State University, Columbus, OH; Electrical & Computer Engineering; M.S., 2006
The Ohio State University, Columbus, OH; Electrical & Computer Engineering; Ph.D., 2010
The Ohio State University, Col., OH; Computer Science & Eng.; Postdoctoral Researcher, 2010–2012
Arizona State University, Tempe, AZ; School of Life Sciences; Postdoctoral Scholar, 2012–present

(b) Appointments

[label= 9999–9999:,leftmargin=*,itemsep=0pt]**Postdoctoral Scholar**, Arizona State University, Austin, TX **Postdoctoral Researcher**, The Ohio State University, Columbus, OH **Hardware R&D Intern**, National Instruments, Austin, TX **Core Systems Software Developer**, IBM Network Storage, Research Triangle Park, NC **Information Technology Systems Engineer**, CallTech Communications, Columbus, OH **Web Developer**, MegaLinx Communications, Dublin, OH

(c) Products

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2012–2013: **1997–2003:** **Pavlic, T.P.**, S. Wilson, G.P. Kumar, and S. Berman. An enzyme-inspired approach to stochastic allocation of robotic swarms around boundaries. In: *Proceedings of the 16th International Symposium on Robotics Research*, December 16–19. 2013. Accepted.

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Kumar, G.P., A. Buffin, **T.P. Pavlic**, S.C. Pratt, and S.M. Berman. A stochastic hybrid system model of collective transport in the desert ant *Aphaenogaster cockerelli*. In: *Proceedings of the 16th International Conference on Hybrid Systems: Communication and Control*, April 8–11, 2013.

Pavlic, T.P. Physical Stigmergy for decentralized constrained optimization: an intelligent lighting example. In: *Proceedings of the 4th International Conference on Cyber-Physical Systems*, April 8–11, 2013. Poster abstract.

Pavlic, T.P., S. P. Peddi, P.A.G. Sivilotti, and B.W. Weide. Getting Out of the Way–Safety Verification without Compromise. In: *Proceedings of the 2012 IEEE/ACM Third International Conference on Cyber-Physical Systems*, April 17–19, 2012. Poster abstract.

Pavlic, T.P., and K.M. Passino. Generalizing foraging theory for analysis and design. *The International Journal of Robotics Research [Special Issue on Stochasticity in Robotics and Bio-Systems Part 1]*. 30(5):505–523. 2011. doi:10.1177/0278364910396551

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Freuler, R.J., M.J. Hoffmann, **T.P. Pavlic**, J.M. Beams, J.P. Radigan, P.K. Dutta, J.T. Demel, and E.D. Justen. Experiences with a Comprehensive Freshman Hands-On Course – Designing, Building, and Testing Small Autonomous Robots. In: *Proceedings of the 2003 American Society for Engineering Education Annual Conference & Exposition*, 2003.

(d) Synergistic Activities

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Conference service: (i) Organizer and associate editor for invited session: "Correctness by Verification and Design", 14th IEEE Conference on Intelligent Transportation Systems, Washington, DC, October 5–7, 2011. (ii) Organizer for proposed mini-symposium: "Optimization and Rationality in Eusocial Insects", 2013 Society for Mathematical Biology Annual Meeting and Conference, Tempe, AZ, June 10–13, 2013 (iii) Co-organizer for technical session: "Complex Systems of Social Insects in Research and Education", 2013 Biomathematics and Ecology: Education and Research, Arlington, VA, October 11–13, 2013.

University service: Chair of committee for the development of biomimicry and bio-inspired research and education initiatives at Arizona State University, 2013.

Community service: (i) As NSF GK-12 fellow, developed and implemented science lessons for local inner-city public school, 2006–2007. (ii) Recognized in *Wired* magazine for OSS work Linux Virtual Server project, 2000.

Mentoring: (i) Founding faculty co-adviser for Interdisciplinary Complexity Science Student Organization at Arizona State University, 2013. (ii) Founding mentor of 2002 FIRST high-school robotics team that received regional honor for inspiration in engineering, Dublin, OH. (iii) Founding mentor of 2001 FIRST LEGO League all-girls middle-school robotics team, Columbus, OH

Teaching: (i) Authored hundreds of pages of Creative-Commons-licensed course material for courses in analog electronics and control for undergraduate and graduate students, The Ohio State University, 2007–2009. (ii) Developed and implemented Electrical and Computer Engineering undergraduate student design project on retrofittable vehicle-to-vehicle communications system for adaptive cruise control in mixed-traffic environments, The Ohio State University, 2012. (iii) Mathematical modeling guest lecturer, Research Strategies for Animal Behavior, Arizona State University, 2013.

(e) Collaborators & Other Affiliations

Collaborators and Co-Editors: *The Ohio State University*: Manas Agrawal; Emrah Adamey; Vijay Gadepally; Arda Kurt; Ashok K. Krishnamurthy; Ümit Özgüner; Jaeyong Park; Kevin M. Passino; Sai Prathyusha Peddi; Keith A. Redmill; Paolo A. G. Sivilotti; Michael Vernier; Alan D. Weide; Bruce W. Weide; *Arizona State University*: Dieter Armbruster; Spring Berman; Aurélie

Buffin; Hinsby Cadillo-Quiroz; Luis Cisneros; Paul Davies; Jessica D. Ebie; James J. Elser; Georgios E. Fainekos; Jennifer Fewell; Jon F. Harrison; Christal Johnson; Yun Kang; Yang Kuang; Ganesh P. Kumar; Alex Nachman; Susan Neuer; Stephen C. Pratt; Hana Putnam; Jason Raymond; P. Logan Rogers; Takao Sasaki; Zachary Shaffer; Oyita Udiani; Taylor Vance; Sara Imari Walker; Sean Wilson; *New Jersey Institute of Technology*: Simon Garnier; Chris Reid

Graduate Advisers and Postdoctoral Sponsors: *The Ohio State University*: Kevin M. Passino; Paolo A. G. Sivilotti; *Arizona State University*: Stephen C. Pratt (Arizona State University)

Thesis Adviser and Postgraduate-Scholar Sponsor: *Undergraduate honors thesis students:* Christal Johnson (Arizona State University). Total graduate students: 0; Total postgraduate scholars: 0.